

Improving Cyclability and Structural Stability of Co-Free Layered Cathode by Controlling Porosity and Cracks in Secondary Particles for Low-Cost and High-Energy LIBs

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The Co-free $\text{Li}[\text{Ni}_{0.75}\text{Mn}_{0.25}]\text{O}_2$ (NM7525) cathode offers the significant advantage of low production costs. However, it requires a high-voltage charging process (≥ 4.5 V vs Li/Li^+) to achieve the energy density level of high-Ni $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ ($x \geq 0.8$) cathodes, which leads to severe structural and morphological degradation in the secondary particles during prolonged cycling. Herein, it is demonstrated that the formation of a stable and homogeneous cathode-electrolyte interface (CEI) can effectively suppress large porosity and crack propagation in the secondary particles of the NM7525 cathode, as well as undesirable structural changes and microstrain in the crystal structure, with a high charging cut-off voltage of 4.45 V and an elevated temperature of 45 °C in the full-cell system. For stabilization of the CEI layer at the high-voltage operation, an optimized electrolyte system is applied, containing the additive with low highest occupied molecular orbital (HOMO) energy. This controlled porosity and crack formation in secondary particles enhances the electrochemical performances of the NM7525-based full cell. In the case of cyclability, the CEI-stabilized full-cell delivers a high-capacity retention of 90% after 100 cycles under the harsh operation conditions of high charging cut-off voltage and elevated temperature, whereas the as-prepared full-cell shows a retention of just 78%.

In the past decade, the typical layered oxide cathodes, such as $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ (NCM) and $\text{Li}[\text{Ni}_x\text{Co}_z\text{Al}_y]\text{O}_2$ (NCA), have been intensively utilized to achieve the required performance on LIBs.^[5–7] Especially, to maximize the specific capacity of the cathode materials, the most recent studies have focused on high-Ni NCM cathode materials ($\text{Ni} \geq 80\%$), which delivered over 800 Wh kg^{-1} for high-energy density.^[8] Nevertheless, the scarcity and price volatility in raw materials, particularly for Co resources, make it difficult for EVs to achieve price parity similar to internal combustion engine vehicles (ICEVs). Given the various global supply chain challenges, conventional high-Ni NCM cathode materials are inappropriate for the economically viable strategy for the EV market. Therefore, the development of eliminating Co in cathode materials for price-neutral LIBs has become a significant research priority.

Recently, Co-free $\text{Li}[\text{Ni}_x\text{Mn}_y]\text{O}_2$ ($x < 0.8$, $y > 0.2$), have been significantly considered the most promising candidates for

next-generation LIBs with high-energy density and low-cost advantage.^[9–11] However, to obtain a high energy density similar to the high-Ni NCM level, an increase of charging cut-off voltage to 4.5 V (vs Li/Li^+) should be required, which results in severe structural change, such as H2–H3 phase transition. Especially, in the case of the secondary-particle-based particles of the layered oxide cathodes, it was reported that the severe structural change induced by the high voltage charging process generates the catastrophic mechanical failure interior to the particles

1. Introduction

With global decarbonization policies and the threat of global warming, the energy paradigm for mobility is shifting toward electrified transportation supported by sustainable energy resources.^[1–3] For the expansion of the electric vehicles (EVs) market and the demand for enhanced driving ranges, the continuous research into Li-ion batteries for high-energy density has necessitated the replacement of engine-based vehicles.^[4]

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and aggravates the particle pulverization.^[12–14] Thus, the high-voltage charging process, essential for achieving the high energy density of $\text{Li}[\text{Ni}_x\text{Mn}_y]\text{O}_2$ cathodes, can induce severe intergranular cracks along the primary particle boundaries within the secondary particles. This compromises the mechanical integrity of the material and leads to the propagation of large cracks in the secondary particles, accompanied by significant internal porosity evolution during prolonged cycling.^[15] Then, the generated cracks and fractures in the secondary particles give rise to exposure of the cathode surface, and electrolyte infiltration contributes to capacity retention fading. Moreover, the formation of cracks in the secondary particles and the degradation of cycle performance, which occurred during the high voltage charging process, is further accelerated under a high temperature (45 °C) environment.^[16] In the case of conventional NCM cathodes, various studies have been conducted to prevent crack formation in secondary particles.^[17,18] However, these studies have primarily focused on improving particle stability through the control of secondary particle formation and the regulation of compositional gradients in these particles. Moreover, since NCM cathodes typically operate with a charging cut-off voltage of 4.3 V (vs Li/Li^+), there remains a lack of intensive research aimed at suppressing crack formation in secondary particles during high voltage operations, such as at more than 4.5 V (vs Li/Li^+), as with $\text{Li}[\text{Ni}_x\text{Mn}_y]\text{O}_2$ cathodes. Therefore, finding effective solutions to address this issue for stable operation of $\text{Li}[\text{Ni}_x\text{Mn}_y]\text{O}_2$ -based LIBs under the high voltage ($\geq 4.5\text{V}$) charging process is urgently needed.

In this study, we focused on the idea that stabilizing primary particles is preferentially important for the overall stabilization of secondary particles during charge/discharge. It was reported that forming a stable cathode-electrolyte interphase (CEI) layer is one of the most essential methods for stabilizing the crystal structure of the cathode materials.^[19–21] The CEI layer is a solid interfacial layer that forms between the cathode and the electrolyte in LIBs. It naturally develops due to electrolyte decomposition on the cathode surface and plays a crucial role in electrochemical performances. The high-voltage operation may deteriorate the surface chemistry of the cathode and the oxidation stability of the electrolyte components with the generation of HF from the LiPF_6 hydrolysis. These by-products from the electrolyte decomposition contribute to the formation of an excessively thick and unstable CEI layer due to irregular deposition with the limited charge transfer of Li^+ . During the prolonged cycling, the inhomogeneous CEI layer undergoes localized de/lithiation, failing to prevent undesired reactions on the cathode surface, ultimately leading to intergranular cracks.^[22] To address these challenges, optimizing a protective and homogeneous CEI layer via the incorporation of an electrolyte additive with high oxidation stability on the cathodes is essential for preserving cathode integrity and enhancing the electrochemical performance for long-term cycling at the high-voltage operation. Thus, we expected that stabilization of the CEI layer in the primary particles under harsh conditions, such as the high voltage charging process above 4.5 V (vs Li/Li^+), can successfully suppress the crack formation in the secondary particles of $\text{Li}[\text{Ni}_x\text{Mn}_y]\text{O}_2$ cathodes even after prolonged cycling. For the formation of a stable and homogeneous CEI layer on the primary particles of the Co-free $\text{Li}[\text{Ni}_{0.75}\text{Mn}_{0.25}]\text{O}_2$ (NM7525) cathode under the high voltage charging process, we used an electrolyte additive, 1,4-butane sultone (BS) having low highest occu-

pied molecular orbital (HOMO) energy compared to other additives used for stabilization of the high-Ni NCM cathodes based on the charging process of 4.3 V (vs Li/Li^+). The homogenous and stable formation of the CEI layer on the primary particles of the NM7525 cathode by application of the BS-containing electrolyte was verified through the ex situ transmission electron microscopy (TEM) analysis.

Interestingly, we demonstrated that stabilizing the CEI layer can successfully suppress porosity change and crack formation in the secondary particles of the NM7525 cathode under the high voltage charging process and elevated temperatures. Especially, through 3D electron tomography based on focused ion beam-equipped scanning electron microscopy (FIB-SEM) serial section technique, we successfully investigated the porosity change and the pore distribution in secondary particles of NM7525 cathode after 100 cycles.^[23–26] It was verified that the application of the stabilization of the CEI layer results in the well-retained porosity in secondary particles of NM7525 cathode even after 100 cycles under the full cell system with the 4.45V-charging process at 45 °C operation, which is clearly distinct from the as-prepared full cell results. Moreover, it was observed that the stabilization of the CEI layer enables notable suppression of crack formation in secondary particles of the NM7525 cathode after prolonged cycling. This well-controlled stability of macro-structure in secondary particles of NM7525 cathode by the stabilized CEI layer enables enhanced electrochemical performances. In the case of the CEI-stabilized full cell (C-S full cell), after 100 cycles with the 4.45V-charging process at 45 °C, its discharge capacity was retained to $\approx 90\%$ of the initial capacity, whereas the as-prepared full cell (A-P full cell) just delivered the capacity retention of $\approx 78\%$. Furthermore, it was observed that the stabilization of secondary particles enables the enhanced power-capability of the NM7525-based full cell with a high energy density.

2. Results and Discussion

2.1. Enhanced Electrochemical Performances of $\text{Li}[\text{Ni}_{0.75}\text{Mn}_{0.25}]\text{O}_2$ -Based Full Cell

Figure 1a,b exhibit the refined X-ray diffraction (XRD) pattern and crystal structure of $\text{Li}[\text{Ni}_{0.75}\text{Mn}_{0.25}]\text{O}_2$ (NM7525) cathode. All the diffracted peaks were well indexed to the typical O3-type layered structure with $R\bar{3}m$ space group. Moreover, in the XRD pattern of the NM7525 cathode, no diffraction peaks related to noticeable impurities or secondary phases were observed, indicating the successful preparation of the NM7525 cathode. The detailed structural information obtained by Rietveld refinement is tabulated in Table S1 (Supporting Information). The low values of the reliability factors ($R_p = 6.87\%$, $R_f = 2.86\%$, and $\chi^2 = 2.74\%$) indicate high accuracy of the refinement. These results also show that Li atoms are situated at $3a$ sites; Ni and Mn atoms randomly occupy at $3b$ sites; O atoms are located at $6c$ sites. The crystal structure of the NM7525 cathode is also confirmed through high-resolution transmission electron microscopy (HRTEM) analyses. Through the HRTEM-based fast Fourier transform (FFT) analyses, it was verified that the interplanar spacing between neighboring fringes was ≈ 0.214 nm, which is consistent with the d -spacing of the (104) plane in the NM7525 structure (Figure 1c and inset). Moreover, the selected area electron diffraction (SAED)

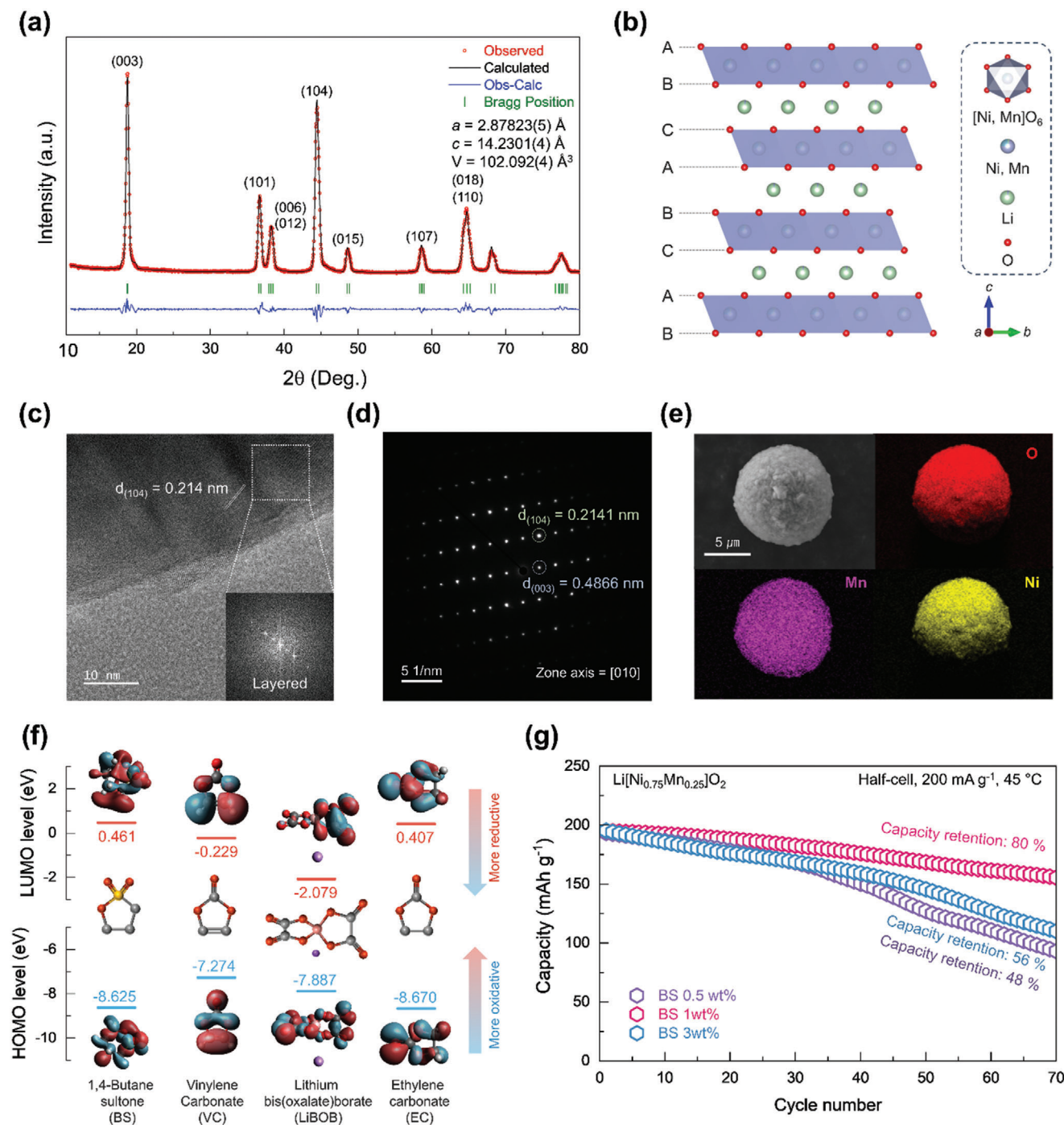


Figure 1. a) Rietveld refined XRD pattern ($R_f = 3.58\%$, $R_p = 2.86$, and $\chi^2 = 2.74\%$) and b) crystal structure of $\text{Li}[\text{Ni}_{0.75}\text{Mn}_{0.25}]\text{O}_2$ (NM7525). c) HRTEM image of NM7525 and corresponding FFT image. d) SAED patterns of as-prepared NM7525 particles e) SEM-EDS mapping images of O, Mn, and Ni elements in NM7525 particles. f) HOMO and LUMO energy level diagrams of electrolyte components. g) Cycle performances of NM7525||Li half cells at 200 mA g^{-1} using various contents of BS additive.

analysis indicates that NM7525 has a single-crystalline structure, in which the obtained structural information is well matched with the Rietveld refinement results (Figure 1d). In Figure S1 (Supporting Information), it was verified that the morphological structure of the NM7525 cathode consisted of cluster-like spherical secondary particles with a diameter range of 5–15 μm via

scanning electron microscopy (SEM) analysis. The SEM based elemental energy dispersive X-ray spectroscopy (EDS) mapping analyses on the NM7525 cathode (Figure 1e; Table S2, Supporting Information) indicate that the Ni and Mn elements are homogeneously distributed in the NM7525 particles, which is also confirmed by the atomic ratio of Ni: Mn = 75: 25 through the

inductively coupled plasma (ICP) analyses (Table S3, Supporting Information).

In addition, the suitability of the BS additive for high-voltage operation was confirmed through density functional theory (DFT) calculations. In terms of the Co-free NM7525 cathode, high-voltage operation above 4.5 V (vs Li/Li⁺) is essential to achieve an energy density comparable to that of Ni-rich NCM cathodes. It was reported that sulfur-containing additives have been utilized for forming a stable CEI layer owing to their inherent high oxidation state (+VI) of the sulfur atom, originating from the decomposition products at high voltage ranges (>4.5 V vs Li/Li⁺).^[27] Moreover, the decomposition products from sulfur-containing additives, such as Li₂SO₄ and ROSO₃Li, can structurally stabilize Ni²⁺ due to their moderately strong affinity for Ni²⁺, thereby effectively suppressing the dissolution of transition metal (TM) ions in Ni-rich cathodes.^[28] Therefore, for facilitating stable CEI formation in the primary particles of NM7525 cathode under high-voltage conditions, we focused on identifying a sulfur-containing electrolyte additive with a low highest occupied molecular orbital (HOMO) energy. As shown in Figure 1f, we calculated and compared the theoretical HOMO and lowest unoccupied molecular orbital (LUMO) of ethylene carbonate (EC), a representative electrolyte solvent, with those of well-known additives vinylene carbonate (VC) and lithium bis(oxalate)borate (LiBOB). The calculation results showed that BS with S = O functional group has a HOMO energy level lower than that of VC and LiBOB but higher than that of EC. This implies that BS participates in reactions to form the CEI layer at higher voltage regions compared to VC and LiBOB, which are typically used for stable operation at 4.3V (vs Li/Li⁺). To experimentally verify this, we evaluated the electrochemical properties using a half-cell system with a charging cut-off voltage of 4.5V (vs Li/Li⁺) and an operation temperature of 45 °C. As shown in Figure 1g, we compared the cycle performance by varying the content of BS and found that the most stable cyclability was achieved with the electrolyte containing 1 wt% BS. Particularly, in the half-cell system, the use of 1 wt% BS-containing (1-BS) electrolyte significantly improved the cycle performance compared to the use of the pristine electrolyte, which implies that the application of 1-BS electrolyte can maximize the stabilization of the CEI layer in NM7525 cathode. To analyze this more accurately, we measured the characteristics in a full-cell system using a graphite anode.

To explore the protective effects of NM7525 cathode by stabilization of CEI layer under harsh conditions in the full cell system, we compared the cycle performances of NM7525||graphite full cells with the pristine and 1-BS electrolytes at a current density of 200 mA g⁻¹ (1C = 276.5 mA g⁻¹) in a voltage range of 2.9–4.45 V at 45 °C. As shown in Figure 2a, the capacity retention of the C-S full cell by use of 1-BS electrolyte was as high as 90% with almost no visible capacity decrease, whereas a gradual capacity fading was apparent for the as-prepared full cell A-P full cell with a capacity retention of 73% after 100 cycles. The drastic capacity decay of the cell with the pristine electrolyte was related to the degradation of the unstable electrode-electrolyte interfaces over 100 cycles. Furthermore, the normalized charge/discharge profiles for the two electrolytes provided a detailed insight into the difference of the cycle performance based on the evolution of the cell polarization during cycling (Figure 2b,c). In terms of the A-P full cell, it was evident that

the normalized capacity was continuously and rather severely reduced during the charge/discharge process up to 100 cycles. In contrast, the C-S full cell showed lower cell polarization than that of A-P full cell, with only a slight capacity fading in the high voltage region. In addition, the X-ray photoelectron spectroscopy (XPS) analysis was conducted to examine the CEI-stabilization effect on the surface of NM7525 cathodes after 100 cycles (Figure S2, Supporting Information). In the C 1s XPS spectra, the peaks corresponding to C–O containing species at 286.5 eV and C–F containing species at 289.8 eV exhibited lower intensity on the cathode in the C-S full cells after 100 cycles, suggesting that the CEI layer is thinner and contains fewer carbonate byproducts from the decomposition of the electrolyte.^[29,30] In the F 1s XPS spectra, the NM7525 cathode in the C-S full cells after 100 cycles showed a less intense Li_xF_yPO_z peak at 686.6 eV compared to that in the A-P full cell. This suggested that the formation of the stable BS-based CEI effectively suppressed electrolyte decomposition during the prolonged cycling under harsh conditions, thereby inhibiting the generation of the Li_xF_yPO_z byproduct.^[31–33] To further demonstrate the CEI components on the surface of NM7525 cathodes via S 2p XPS spectra, it was verified that the decomposition products of BS additive, such as Li₂SO₃ and ROSO₂Li in the 100-cycled cathode electrodes in the C-S full cells. The abundant Li₂SO₃ and ROSO₂Li would participate in a stable CEI layer in the high-voltage operation upon the prolonged cycling (Figure S3, Supporting Information).^[34] Furthermore, the composition of CEI layers was investigated via attenuated total reflection-Fourier transform infrared spectroscopy (ATR-FTIR) to verify the correlation between the surface chemistry of the NM7525 cathodes and the BS incorporation. Figure S4 (Supporting Information) exhibited the ATR-FTIR spectra in the 100-cycled electrode in the C-S full cell. Consistent with the XPS results, the peaks between 775 and 830 cm⁻¹ in the 100-cycled C-S full cell reflected the CEI structure indicating the presence of LiF, HCOOLi, and (CH₂OCOOLi)₂, which are associated with the decomposition of electrolyte on the cathode surface.^[31] Moreover, a peak at 668 cm⁻¹, related to Li₂SO₃, suggested that the BS additive participated in the formation of the CEI layer.^[35,36] In addition, to verify the probability of universal applicability, the full-cell tests of Li[Ni_{0.8}Co_{0.1}Mn_{0.1}]O₂ (NCM811) cathodes were performed under the identical conditions of NM7525-based full-cell tests (Figure S5, Supporting Information). The C-S NCM811-based full cell with the BS-added electrolyte exhibited a capacity retention of ≈73% for 100 cycles, while A-P NCM811-based full cell using the pristine electrolyte just delivered a capacity retention of ≈64%. These results indicate that stabilizing the CEI layer in the primary particles of Ni-rich layered oxide cathodes helps control porosity and minimize crack formation in the secondary particles, thereby enhancing the electrochemical performance of both NM7525 and NCM811 cathodes under harsh conditions, including high-voltage operation at 4.45 V and an elevated temperature of 45 °C.

To further highlight the performance difference, the comparison of the average discharge voltages was obtained from the voltage profiles for the C-S and A-P full cells during 100 cycles (Figure 2d). The A-P full cell exhibited the average discharge voltage decline at a relatively steep rate of 0.62 mV per cycle, whereas the C-S full cell revealed a more gradual decrease of 0.45 mV per cycle. It was associated with a notably specific energy loss

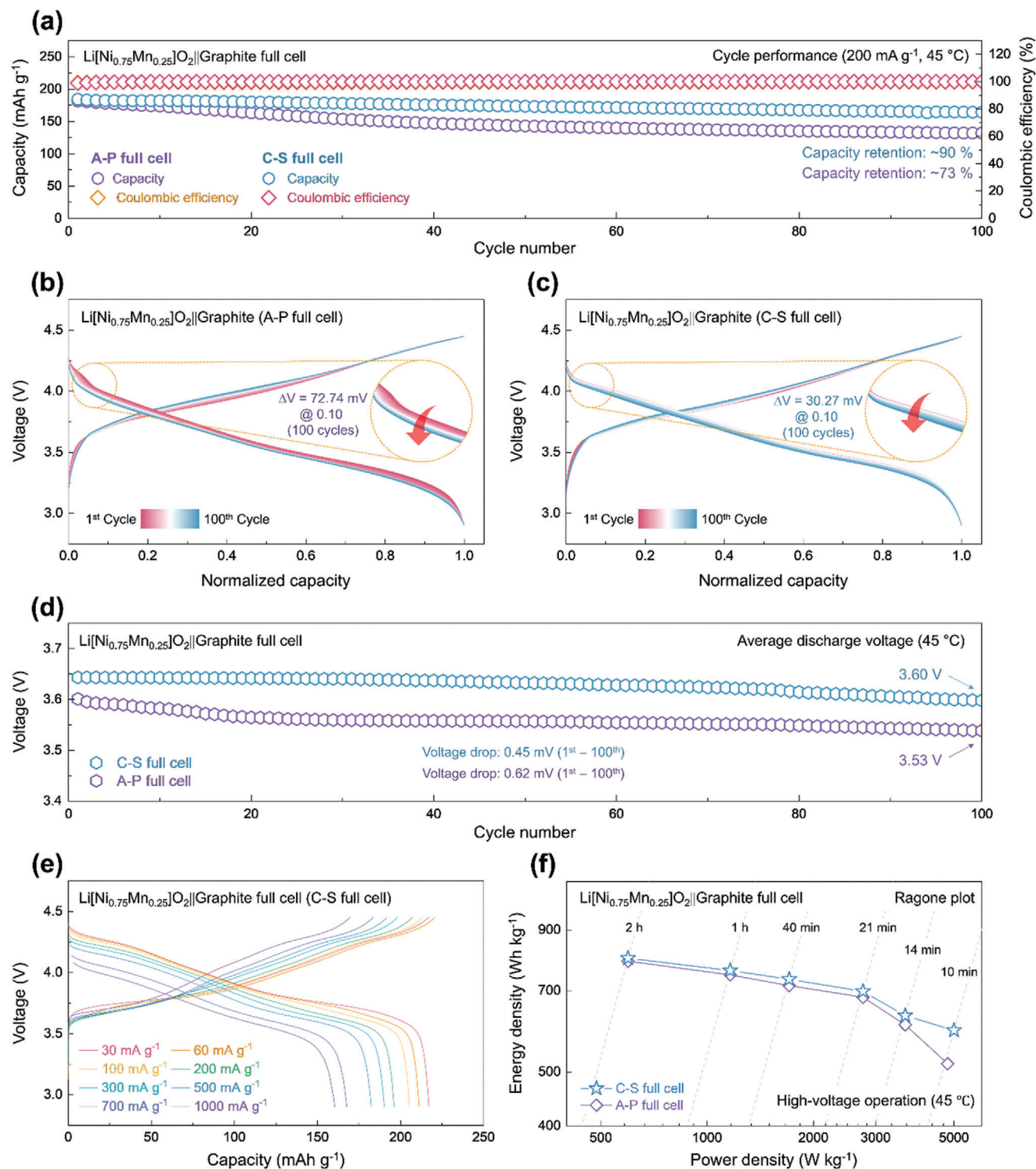


Figure 2. a) Cycle-performances of the as-prepared (A-P) and CEI-stabilized (C-S) full cells under a charge cut-off voltage 4.45 V. Comparison of normalized capacity charge/discharge profiles for b) the as-prepared (A-P) full cell and c) the CEI-stabilized (C-S) full cell during 100 cycles. d) Comparison of average discharge voltages in the as-prepared (A-P) and CEI-stabilized (C-S) full cells. e) Charge/discharge profiles of the CEI-stabilized (C-S) full cell at various current densities and a temperature of 45 °C. f) Ragone plot of the as-prepared (A-P) and CEI-stabilized (C-S) full cells at various current densities.

stored in the cell with the pristine electrolyte, originating from severe degradation of the NM7525 structure for prolonged cycles. Thus, the above results indicated that the cycling stability of the NM7525-based full cell can be highly improved through the stable CEI formation under high-voltage operation. To investigate the difference of the capacity and voltage fading in the 100-cycled NM7525 cathode electrodes, the differential capacity versus voltage (dQ/dV) profiles were compared in Figure S6 (Supporting Information). The electrochemical performance fading is certainly associated with an undesirable phase transition occurring in the high-voltage region. As shown in Figure S6a (Supporting Information), the dQ/dV profiles for the monoclinic phase in the range of 3.6 V to 3.9 V were noticeably invisible during 100 cycles in the A-P full cell. In contrast, the C-S full cell preserved the monoclinic phase with the stable and smooth phase transition after 100 cycles by the homogeneous formation of a stable CEI layer (Figure S6b, Supporting Information). These findings in the pristine electrolyte implied that the phase transition without the monoclinic phase is caused by the severe structural degradation accumulated after 100 cycles, eventually resulting in incremental microcracking at the grain boundaries of the primary cathode particles. Furthermore, it is clearly seen from Tables S4 and S5 (Supporting Information) that the proportion of dQ/dV curves between 4.1–4.45 V distinctly increased substantially after 100 cycles in the A-P full cell (+2.43%) compared to that in the C-S full cell (+1.57%). Since the structural stability of the NM7525 cathode upon prolonged cycling was affected by the mechanical strain related to detrimental H2–H3 phase transition, we assumed that the stabilization of CEI layer effectively mitigated the deteriorated phase transition in the NM7525 structure even after charging to 4.45 V at 45 °C. The incorporation of BS additive is proposed to contribute to the reconstruction of the robust CEI layer on the NM7525 cathode, which is expected to exhibit enhanced electrochemical stability. To assess the effect of Li^+ kinetics through the CEI layer, cyclic voltammetry (CV) measurements were conducted at various scan rates from 0.1 to 0.5 mV s^{-1} in the voltage range of 2.90–4.45 V. Figure S7a,b (Supporting Information) exhibited that for the first anodic current branch, three obvious peaks observed at 3.55 V, 3.72 V, and 4.25 V are attributed to $\text{H1} \rightarrow \text{M}$, $\text{M} \rightarrow \text{H2}$, and $\text{H2} \rightarrow \text{H3}$ phase transitions, respectively, in both the C-S and A-P full cells. Additionally, the analysis of the CV profiles at various scan rates enabled the assessment of the impact of the surface pseudo-capacitance on lithium storage kinetics.^[37] The relationship between the peak current (i) and scan rate (ν) was analyzed using a function of $i = a\nu^b$ to investigate the charge storage process (Figure S7c, Supporting Information). The b -value for the C-S full cell was calculated to 0.77, higher than that of the A-P full cell, suggesting that the lithium storage mechanism is predominantly dominated by the pseudo-capacitance behavior.^[38] In addition, it is important to confirm that the electrolyte additive successfully participates in the CEI formation during charge/discharge. An electrolyte additive that does not properly react during charging can increase resistance, hindering Li^+ diffusion. To verify the participation of BS additive for the CEI formation, we conducted the Galvanostatic intermittent titration technique (GITT) under a low current density (30 mA g^{-1}) after the initial pre-cycle at the high temperature of 45 °C. The GITT profiles and the apparent Li^+ diffusivity (D) of the C-S full cell and A-P full cell were displayed in Figure

S8 (Supporting Information). The diffusion coefficient of Li^+ ions (D) during charge/discharge was calculated using the following Equation 1:

$$D = \frac{4}{\pi\tau} \left(\frac{mV}{MS} \right)^2 \left(\frac{\Delta E_s}{\Delta E_t} \right)^2, \left(\tau \ll \frac{L^2}{D} \right) \quad (1)$$

where τ , ΔE_t , and ΔE_s represent the duration of constant current, the voltage variation during a single step, and the cumulative voltage change under applied current conditions, respectively. The diffusivity of Li^+ ions for the C-S full cell and the A-P full cell after the charging process were evaluated as $8.5 \times 10^{-7} \text{ cm}^2 \text{ s}^{-1}$ and $8.2 \times 10^{-7} \text{ cm}^2 \text{ s}^{-1}$, respectively. It indicates that the incorporation of BS additive facilitated the formation of a stable CEI layer, leading to a modest increase in diffusivity that provides an effective pathway for Li^+ diffusion. Moreover, this finding is well matched with the CV results, which indicated that the decomposition of BS additive in the high-voltage region on the cathode surface effectively induces the formation of a stable CEI without increasing the internal resistance during the charge/discharge process.^[39] Moreover, the electrochemical impedance spectroscopy (EIS) analysis was performed to support the effect of CEI-stabilization in lowering the resistance of 100-cycled full cells at the high temperature of 45 °C (Figure S9, Supporting Information). In the C-S full cell, the overall EIS profile exhibited smaller curves of the cathode-electrolyte interface resistance (R_{CEI}) and charge transfer resistance (R_{CT}) compared with the results on the A-P full cell. These results indicated that the creation of the protective CEI layer contributed to enhancing the interfacial stability of the NM7525 particles under harsh operation conditions.

The stable formation of CEI layer also leads to the enhanced power-capability of the NM7525-based full cell. As shown in Figure 2e and Figure S10 (Supporting Information), the specific capacities of the A-P and C-S full cells measured at a current density of 30 mA g^{-1} were ≈ 215.2 and $\approx 217.1 \text{ mAh g}^{-1}$, respectively. Especially, the C-S full cell at 1000 mA g^{-1} exhibited a specific capacity of $\approx 159.8 \text{ mAh g}^{-1}$, whereas the A-P full cell delivered a specific capacity of $\approx 138.3 \text{ mAh g}^{-1}$ at the same conditions. To confirm the impact of loading mass level on the cathode electrodes, the electrochemical performances of NM7525 cathode with different loading mass were conducted at various current densities (Figure S11, Supporting Information).^[37,40] Through the comparison using the Ragone plot (Figure 2f), the enhanced power-capability by the stable formation of the CEI layer is more clearly revealed, which is attributed to the improved Li^+ diffusion by protecting the interfacial electrodes in harsh conditions.

2.2. Controlled Porosity and Cracks in Secondary Particles Via Stabilization of CEI Layer

Referring to previous research, the LiPF_6 hydrolysis in the electrolyte promotes the formation of acid compounds (e.g., HF), which induces TM dissolution on the cathode surface. It contributes to severe interfacial degradation caused by the H2–H3 phase transition, ultimately leading to the microcrack formation between primary particles. Especially as structural degradation of Ni-rich cathode materials occurs at the CEI layer due to the deteriorated reaction between highly reactive Ni^{4+} and the electrolyte,

the microcracks affect the increase of the internal area exposed to the electrolyte, leading to the electrolyte infiltration in the cathode particle interior. The subsequent degradation of the exposed contact area due to parasitic electrolyte attack accelerates the fracture of the cathode's secondary particles and aggravates capacity fading during prolonged cycling. To solve these problems, the formation of a stable and robust CEI layer on the cathode electrode has been continuously conducted to alleviate the severe structural degradation during prolonged cycling.^[21] The enhanced cycling performance of the C-S full cell is attributed to the formation of the stable and homogeneous CEI layer, which protects the undesirable electrolyte penetration and inhibits structural degradation. To further assess the effect of the stabilized CEI layer on the cathode particles in detail, we investigated the contrasting inner structural changes in the NM7525 secondary particles between the A-P and C-S full cells in the harsh conditions (high charging cut-off voltage 4.45 V and elevated temperature 45 °C).

To characterize internal pore distribution structures from a 3D perspective, the electron tomography technique was employed based on the serial sectioning method using FIB-SEM. This involves collecting a series of 2D SEM images during the FIB serial sectioning, which can subsequently be aligned to construct a 3D volume representing the sample under investigation.^[24] In SEM imaging, secondary electron (SE) imaging is favored for visualizing surface morphology due to its high contrast at object edges, resulting from increased secondary electron emission. However, this SE imaging can lower the reliability of object recognition and segmentation, which are essential processes for electron tomographic analysis.^[41] Conversely, backscattered electron (BSE) imaging, which collects recoiled electrons after elastic beam-sample interactions, shows contrast dependent on atomic number, unaffected by surface geometry. Since the BSE imaging is thus free of topographic contrast changes, a series of BSE images during the FIB serial slicing of the sample were acquired for a reliable object segmentation for the tomographic volume reconstruction (Figure S12, Supporting Information). This FIB-SEM-based serial section imaging technique for artificial volume reconstruction is highly effective for visualizing and analyzing the 3D structure of a hierarchically structured secondary particle that bears primary grains and geometrical imperfections such as internal pores (or cracks). Additionally, it offers a superior spatial resolution (a few nanometers) that surpasses the capabilities of X-ray computed tomography, which is vital for the 3D examination of nanoscale structural alterations.^[42]

Figure 3a describes the typical process of a pore-segmented volume reconstruction for the NM7525 cathode particles using the above-mentioned FIB-SEM-based serial section imaging. In the FIB-SEM instrument, the focused Ga ion beam was precisely aligned to the desired sectioning plane in the sample and then continuously sliced off the sectioning plane at the depth interval of 15 nm, while the SEM was synchronized to acquire every exposed sectioning plane automatically. This FIB-SEM serial sectioning process generated a stack of SEM images representing the internal structure of the sample. Through the binarized segmentation process that automatically groups pixels into subsets for separating semantic regions or objects in SEM images, an artificial image stack divided by pores (and cracks) and filled particle regions was generated, ready for 3D volume reconstruction for visualization and quantification of pore distribution inside

the analyzed particles.^[43] Figure 3b presents the reconstructed volume (left panel) and independently separated internal pore structure (right panel) of the fresh NM7525 secondary particle before cycling. The reconstructed tomographic volume showed that the NM7525 secondary particle bears two distinctive porous structures: a spherical porous layer decorating the outskirt of the core part and large pores clustered in the core region. Likewise, a similar geometrical distribution of pores was observed in the reconstructed tomographic volumes for the 100-cycled NM7525 secondary particles with or without stabilization of the CEI layer (Figure 3c,d). Nonetheless, it is observed that the sizes and morphological distribution of internal pores were developed differently depending on whether stabilization of the CEI layer. The live rotation and perspective views of the three tomographic volumes with the segmented pore structures are provided in supporting Videos S1–S3 (Supporting Information), respectively.

To evaluate the difference in pore structure evolution after 100 cycles by stabilization of the CEI layer, we calculated pore contents (i.e., total porosity, P_t) and the total surface area (S_t) of the pores inside the reconstructed volumes, as shown in Figure 3e,f. The P_t of the secondary particle in the 100-cycled C-S full cell was evaluated to be $\approx 6.14\%$, which is similar to that of the fresh secondary particle ($\approx 5.88\%$). In contrast, the secondary particle in the 100-cycled A-P full cell showed a considerably larger P_t ($\approx 8.43\%$) than that of the fresh secondary particle, which was attributed to the structural deterioration causing the gradual growth of internal pores and crack propagation in the secondary particles under the prolonged high voltage cycles. Moreover, as the internal porosity decreases due to the suppression of cycling-induced structural degradation by stabilization of the CEI layer, the total surface area (S_t) in the secondary particle correspondingly decreases, which is a signature of the suppressed crack propagation. This result demonstrates that the secondary particle of NM7525 in the A-P full cell can be readily damaged upon cycling. In contrast, the uniform CEI distribution in the C-S full cell notably improved electrode stability under harsh high-voltage cycling. Furthermore, EDX elemental mapping for the three samples verified that the C-S full cell delivered effectively suppressed TM dissolution after extended cycling, compared to the A-P full cell (Figure S13 and Table S6, Supporting Information). These results of structural and chemical characterizations suggested that the robust and homogeneous CEI layer effectively restrained TM dissolution from the NM7525 cathode and mitigated the formation of internal pores.

To obtain deeper insight into the enhanced stability in the secondary particles of NM7525 cathode by stabilization of the CEI layer under the harsh electrochemical cycles, direct measuring of the 3D distribution of internal pores is crucial because existing local pores and newly formed cracks become combined into larger pores as the cycle-induced structural pulverization proceeds. The geometrical pore distributions inside the three cathode samples were visualized, as shown in Figure 4a. Note that the positions of internal pores and their size were determined to correspond to the artificial sphere filling for the inside of an irregular shape of an actual pore for simplicity in the digital processing for pore detection.^[24,44] Compared to the initial pore distribution in the fresh particle, significant internal pore evolution was observed in the 100-cycled secondary particle of the NM7525 cathode within the A-P full cell. In contrast, the 100-cycled secondary particle in

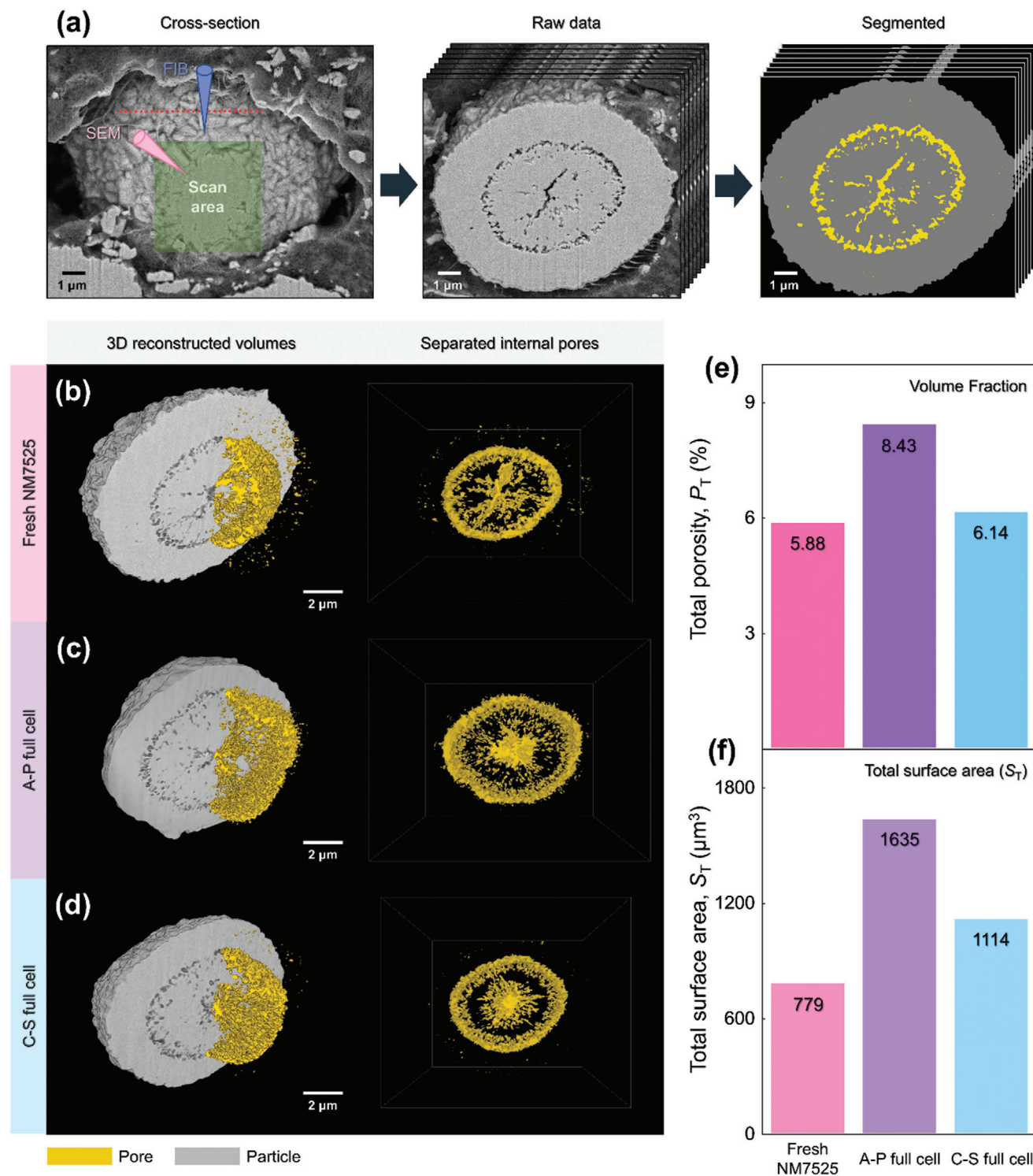


Figure 3. a) Typical workflow of FIB-SEM serial sectioning and digital segmentation of pore structure in the secondary particle to reconstruct a tomographic volume. b–d) Perspective views of 3D reconstructed volumes (left panel) and separated internal pore structures (right panel) of the fresh NM7525 and the 100-cycled electrodes in the as-prepared (A-P) and CEI-stabilized (C-S) full cells, respectively. Quantitative analysis on e) total porosity (P_T) and f) total surface area (S_T) calculated from the reconstructed volumes.

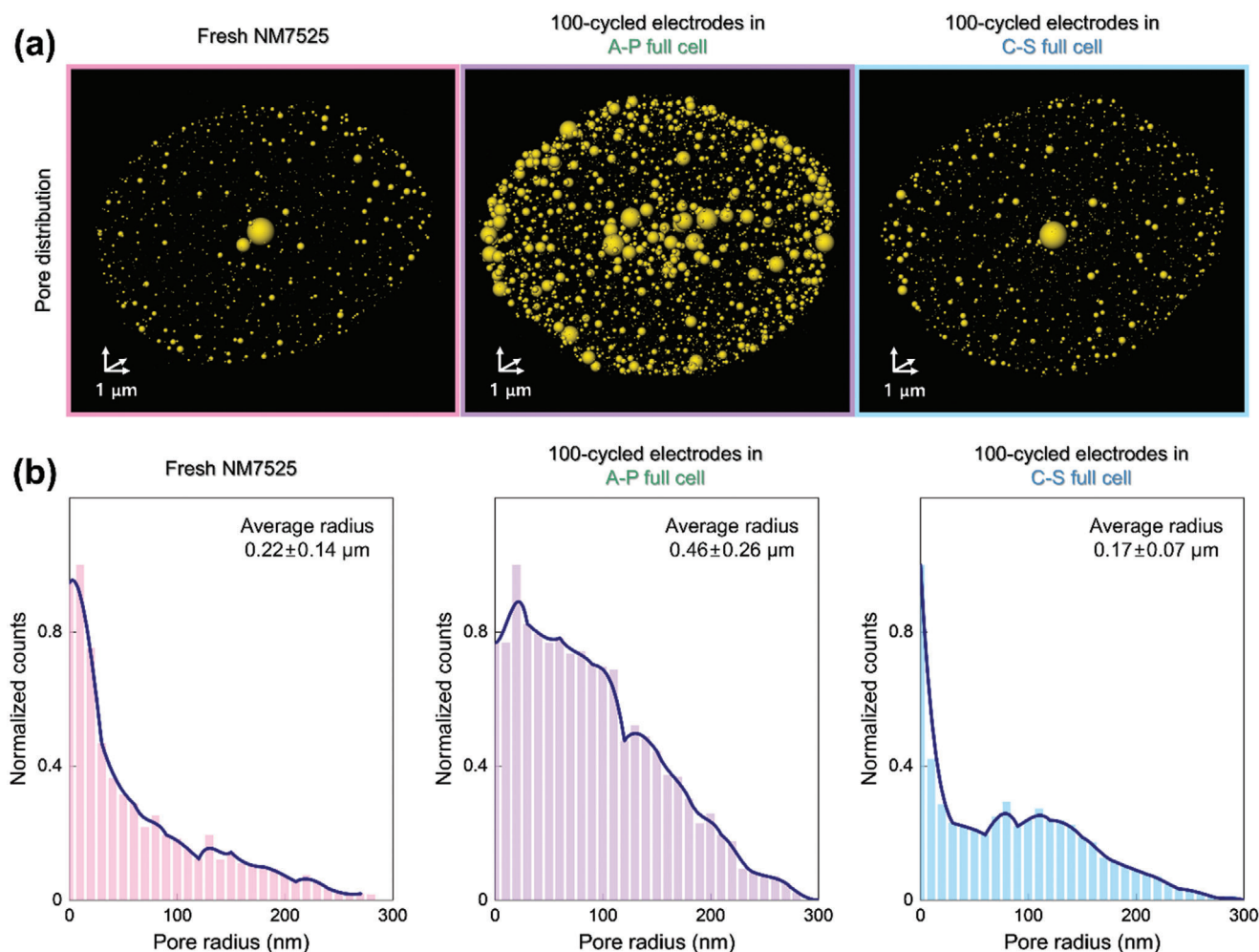


Figure 4. a) 3D distributions of internal pores reconstructed from the tomographic volumes of the fresh NM7525 and the 100-cycled electrodes in the as-prepared (A-P) and CEI-stabilized (C-S) full cells, respectively. b) Histograms showing pore diameter distributions of the three samples.

the C-S full cell exhibited a pore distribution similar to that of the fresh secondary particle. These results suggest that the formation of pores and cracks in the secondary particle of NM7525 during cycling is effectively suppressed by the stabilization of the CEI layer. By quantifying pore size distribution, we notice that the average size and 3D distribution of internal pores in the secondary particle of NM7525 in the C-S full cell are similar to the fresh secondary particle even after 100 cycles. In contrast, the pore sizes within the secondary particles of NM7525 in the A-P full cell became larger, with a broader distribution compared to the other samples, as evident from the histograms of the 3D pore structure distributions (Figure 4b). These results revealed a clear correlation between the degradation of electrochemical properties and the degree of pore structure evolution inside the secondary particles. As the incremental creation of internal pores under harsh electrochemical cycling induces the increase of the exposed surface area of primary particles to the parasitic electrolyte, it eventually leads to more severe degradation of the electrochemical properties for the sample when the CEI layer is not robust to protect it against the electrolyte infiltration into the secondary particle. In this respect, it is obvious that the CEI layer formed on

the NM7525 in the C-S full cell is highly robust and stable at the high voltage cyclic operation, resulting in the suppression of pore formation in the secondary particles. Considering the aforementioned results, along with the fact that the C-S full cell outperforms the A-P full cell in respect of electrochemical properties at the high voltage operation, it was confirmed that the formation of a homogeneous CEI layer as a protective layer on the cathode surface inhibits the growth of various pores in the secondary particles, independent of grain size change, and minimizes the degree of electrolyte propagation by reducing the area exposed to the electrolyte.

In addition, the morphology of the 100-cycled secondary particle in the NM7525 cathode was investigated in detail, using the cross-sectional SEM analyses. As shown in Figure 5a and S14 (Supporting Information), severe crack propagation and degradation were observed in the secondary particles of the 100-cycled A-P full cell. In contrast, even in the elevated temperature and high charging cut-off voltage, the secondary particle in the 100-cycled C-S full cell exhibited effectively suppressed critical cracks on the particles and maintained a morphology similar to the fresh electrode. These results indicate that the deleterious

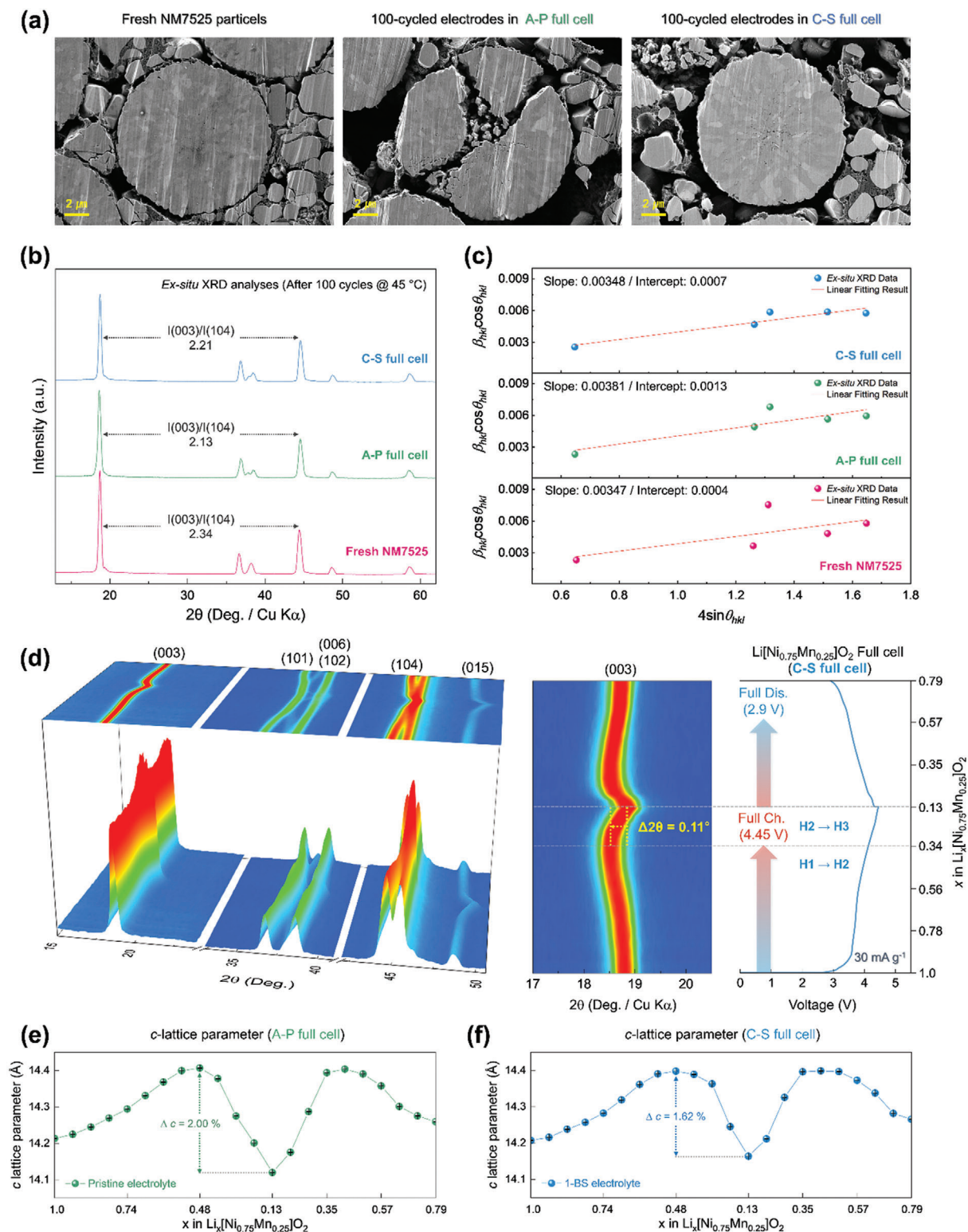


Figure 5. a) Magnified cross-sectional SEM images of fresh NM7525 and the 100 cycled-NM7525 secondary particles in the as-prepared (A-P) and CEI-stabilized (C-S) full cells, respectively. b) Ex situ XRD patterns of the fresh NM7525, the 100-cycled electrodes in the as-prepared (A-P) and CEI-stabilized (C-S) full cells. c) W-HISM plots of NM7525 based on *ex-situ* XRD patterns of fresh NM7525 and 100-cycled electrodes in the as-prepared (A-P) and CEI-stabilized (C-S) full cells, respectively. d) Operando XRD analysis of the CEI-stabilized (C-S) full cell at 30 mA g^{-1} : Overall XRD diagrams (left), magnified contour plot of (003) plane (middle), and the corresponding charge/discharge profile (right). Change in the *c*-axis lattice parameter of NM7525 in (e) the as-prepared (A-P) and (f) the CEI-stabilized (C-S) full cells for the first charge/discharge.

electrochemical performance and structural instability of the NM7525 cathode are particularly attributed to the severe degradation of the particle surface exposed to the electrolyte. Moreover, the XRD analyses of the 100-cycled electrodes in the C-S full cell supported the enhanced structural stability of the NM7525 cathode by stabilized CEI layer for prolonged cycles, as shown in Figure 5b. After 100 cycles, the (003) peak intensity was notably lower than the (104) peak in the A-P full cell, whereas the C-S full cell was well retained without considerable degradation. Considering the ratio of $I_{(003)}/I_{(104)}$ related to the $\text{Li}^+/\text{Ni}^{2+}$ cation mixing in Li layer, the lower value in the A-P full cell indicated that the undesirable cation mixing was experienced after prolonged cycling.^[45,46] In contrast, the stabilization of the CEI layer effectively preserved the crystal structure of the NM7525 cathode with high value. The reason for the higher $I_{(003)}/I_{(104)}$ ratio in the C-S full cell can be attributed to the robust CEI layer securing the electrolyte against the highly reactive Ni^{4+} during the prolonged cycles, alleviating $\text{Li}^+/\text{Ni}^{2+}$ cation mixing and structural degradation.

Especially, to further elucidate the differences of the lattice evolution in NM7525 cathode particles in A-P and C-S full cells over the prolonged cycles, we quantified the lattice microstrain during each cycle via Whilliam-Hall isotropic model (W-HISM).^[47,48] The corrected full width at half maximum (FWHM) of each X-ray diffraction peak for the cathode material was calculated due to the instrumental effect associated with peak broadening using the following Equation 2:

$$\beta_{\text{hkl}} = \sqrt{\beta_{\text{measured}}^2 - \beta_{\text{instrumental}}^2} \quad (2)$$

where β_{hkl} , β_{measured} , and $\beta_{\text{instrumental}}$ represent the corrected, measured, and instrumental peak width of X-ray diffraction peak profile in radians, respectively. The instrumental data was obtained using lanthanum hexaboride (LaB_6) as standard powder. In the W-HISM plot, the corrected β_{hkl} , contributed to both lattice strain and crystallite size. Then, considering the two independent factors of lattice strain and crystallite size in a particular peak, the total peak broadening, related to the isotropic nature of the crystals, was expressed in the following Equation 3:

$$\beta_{\text{hkl}} \cos \theta_{\text{hkl}} = 4 \varepsilon \sin \theta_{\text{hkl}} + \frac{K\lambda}{D} \quad (3)$$

where θ_{hkl} is the Bragg angle, D is the crystallite size, λ is the X-ray wavelength, and K is assumed to be 0.9 as the shape factor. Table S7 (Supporting Information) exhibited the lattice strain (ε) information from the linear fitting with a term of $4 \sin \theta_{\text{hkl}}$ along x -axis and $\beta_{\text{hkl}} \cos \theta_{\text{hkl}}$ along y -axis based on the above equation. As presented in Figure 5c, during the cycling up to 100 cycles, the total lattice strain of NM7525 cathode in the A-P full cell increased by $\approx 11.24\%$ compared to the original value, whereas that of NM7525 in the C-S full cell increased by only $\approx 0.29\%$. The suppressed structural and morphological changes in NM7525 particles by the stabilized CEI layer are also confirmed through ex situ TEM analysis. Figure S15 (Supporting Information) exhibited the TEM images of NM7525 cathodes in the A-P and C-S full cells after 100 cycles at 45°C . In the case of the A-P full cell, the uneven and uncontrolled CEI layer was formed on the surface of NM7525 particles after cycling. Especially, the inhomogeneous

CEI layer exhibited a thickness variation ranging from ≈ 1.30 to 7.02 nm. On the contrary, the thickness of the CEI layer of the NM7525 particle in the cycled C-S full cell is ≈ 3.61 nm, which is thinner and more homogeneous than that of the pristine electrolyte. Furthermore, the HRTEM images and the corresponding fast Fourier transform (FFT) images in Figure S16 (Supporting Information) revealed that the surface and bulk region of NM7525 secondary particle in the A-P full cell after 100 cycles exhibited a rock-salt ($Fm\bar{3}m$) phase with an interplanar distances of 0.22 nm.^[39] It was speculated that the severe structural deterioration in the A-P full cell during cycling occurred under harsh operation conditions (4.45 V and 45°C). In contrast, the C-S full cell after 100 cycles exhibited relatively mitigated structural degradation, as evidenced by the presence of both layered ($R\bar{3}m$) and spinel-like ($Fd\bar{3}m$) phases at the outmost surface, suggesting that TM dissolution was effectively restrained. These results indicate that the robust and homogeneous CEI layer protects the interfacial cathode electrode and suppresses the serious mechanical strain, thereby alleviating the structural degradation of NM7525 secondary particles for prolonged cycling.

To further investigate the structural evolution of NM7525 cathode upon cycling and the effect of the cathode-electrolyte interface chemistry under the high voltage region, *operando* XRD analyses were conducted in the first cycle at 30 mA g^{-1} through the full cell system. Figure 5d and Figure S17 (Supporting Information) exhibited that the (003) reflection of the NM7525 cathode in the A-P full cell shifted noticeably with the state of charge (SoC), indicating a variation along the c -lattice parameter of $R\bar{3}m$ space group. Especially, the (003) reflection shifts toward slightly lower 2θ angles during charging from 3.0 to 4.1 V, indicating the increase of the c -lattice parameter due to the H1-H2 phase transition in NM7525 cathodes. With the subsequent charging to 4.45 V, the (003) reflection reversely moves to higher 2θ angles at a high state of charge, driven by the extraction of most Li^+ ions from the Li-layer, ultimately contracting during the H2-H3 phase transition. Notably, the extent of the shift in the C-S full cell was smaller than that in the A-P full cell, providing highly enhanced structural stability of the NM7525 cathode through stabilization of the CEI layer. To examine the structural difference in more detail, the c -lattice parameter was calculated from the Rietveld refinement based on the *operando* XRD patterns for the entire cycle process of the NM7525 cathode in the A-P and C-S full cell (Figure 5e,f). The c -lattice value exhibited a gradual increase but experienced an abrupt contraction when the deleterious H2-H3 phase transition occurred in the highly delithiated state. During charging from 4.1 to 4.5 V, the degree of c -axis contraction in the C-S full cell ($\approx 1.62\%$) is lower than that in the A-P full cell ($\approx 2.00\%$). These results demonstrate that CEI control effectively suppresses undesirable structural changes occurring during the high-voltage charging process of NM7525, supporting the successfully inhibited formation of large cracks and porosity in the secondary particles of NM7525 during prolonged cycling.

As summarized in Figure 6, for secondary particles of the NM7525 cathode in the A-P full cell, numerous microcracks formed along the intergranular boundaries in a random manner during charge/discharge. These intergranular cracks developed after long-term cycling, creating pathways for undesirable

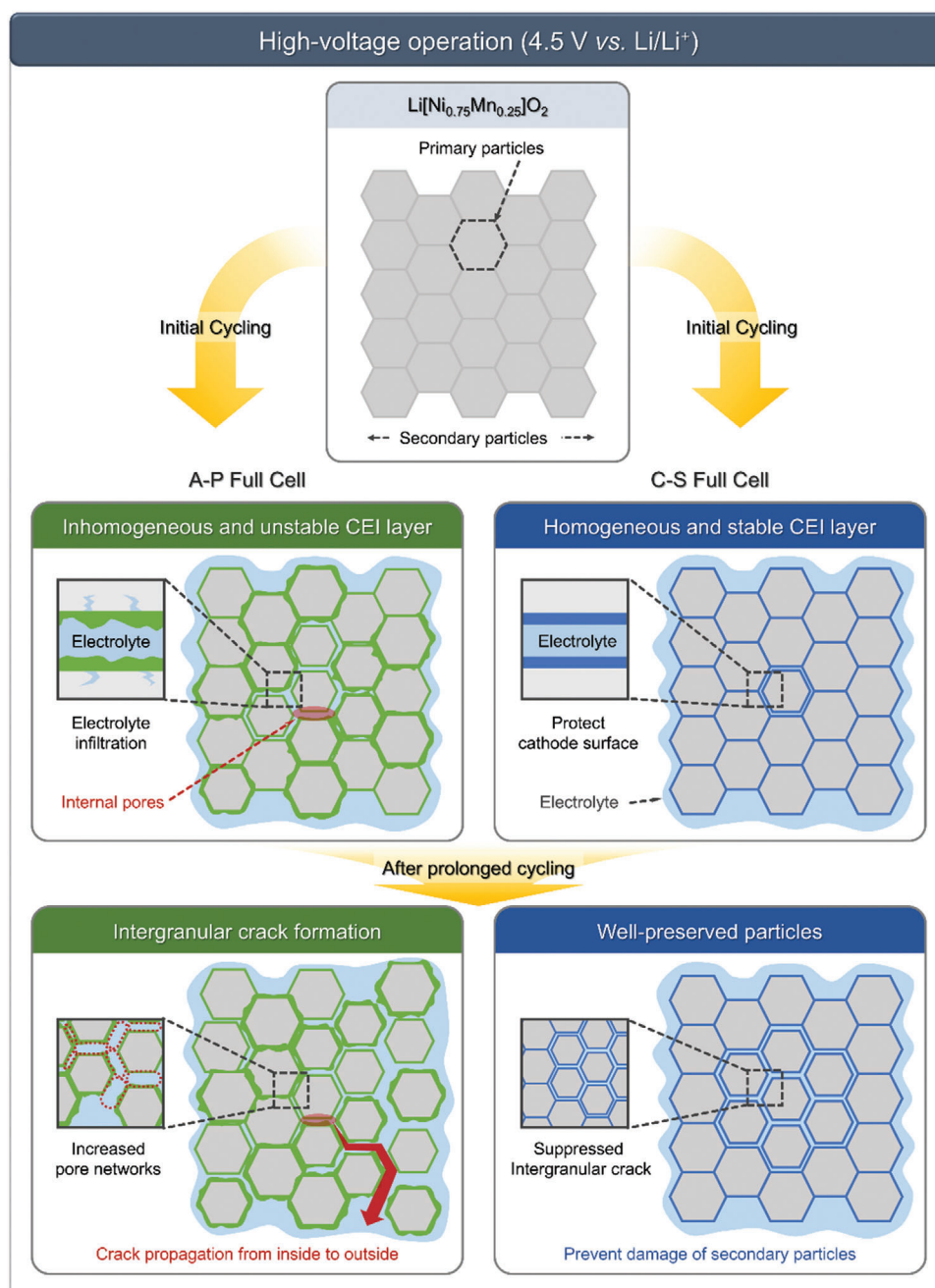


Figure 6. The schematic image on the effect of CEI-stabilization for suppressed porosity and cracks in the NM7525 secondary particles during high-voltage-based cycling.

electrolyte infiltration. The incremental formation of the microcracks further led to the development of an internal pore network, which resulted in morphological degradation and mechanical pulverization in the secondary particles of the NM7525 cathode. On the other hand, in the case of the C-S full cell, the intergranular cracking inside the secondary particle of the NM7525 cathode was effectively reduced with the formation of the homogeneous CEI layer. These findings demonstrate that effectively stabilizing the CEI layer significantly suppresses severe intergranular cracks and large porosity in the secondary particles, protecting them

from parasitic electrolyte side reactions. This stabilization successfully enhances the homogeneous (de)lithiation and electrochemical performance of the NM7525 cathode, even under harsh operating conditions.

3. Conclusion

We demonstrated that the formation of porosity and cracks in the secondary particles of the Co-free NM7525 cathode can be successfully supported through the stabilization of CEI layers

during the high voltage charging process in the full cell system. The C-S full cell exhibited a high capacity retention of 90% after 100 cycles under harsh conditions of the high cut-off voltage (4.45 V) and the high temperature (45 °C), whereas the A-P full cell just delivered a capacity retention of 78% at the same conditions. Especially, through FIB-SEM tomography, it was revealed that the formation of a stable CEI layer by usage of BS additive with low HOMO energy can reduce the surface exposure and mitigated the porosity development and crack propagation in the secondary particles of NM7525 cathode, which clearly distinct from the results based on the pristine electrolyte. To further investigate the cracks in the secondary particle, cross-sectional SEM images of the 100-cycled NM7525 cathode were obtained, proving that the secondary particle of NM7525 cathode in the A-P full cell showed distinct fractures and crack propagation due to an unstable cathode surface compared to that in the C-S full cell. Moreover, by calculating the lattice microstrain during each cycle and conducting *operando* XRD measurements, it was confirmed that the stabilized CEI layers effectively suppress undesirable large structural changes in the NM7525 cathode during charge/discharge with the high charging cut-off voltage and the elevated temperature. This supports the successful inhibition of large crack and porosity formation in the secondary particles of the NM7525 cathode during prolonged cycling. We believe this work offers valuable insights into improving the electrochemical properties of Co-free layered cathodes through an effective strategy for stabilizing the secondary particles of cathode materials, paving the way for low-cost, high-energy LIB.

4. Experimental Section

Preparation of Electrolyte and Electrode: The pristine electrolyte consisted of 1.2 M LiPF₆ dissolved in EC/EMC/DMC (2:4:4 vol%). The BS-containing electrolyte (pristine electrolyte + 1 wt% 1,4-butane sultone (BS, ≥ 99%, Sigma-Aldrich)) was prepared in an Ar-filled glove box. After blending the electrolyte solution, 1 wt% BS-containing (1-BS) electrolyte was stirred at 100 RPM at 60 °C for 60 h and rested for 2 days to stabilize the solution mixture.

For the preparation of NM7525 cathode particles, co-precipitated (Ni_{0.75}Mn_{0.25})OH precursor was mixed with LiOH at a stoichiometric ratio. The NM7525 mixture was calcined at 850 °C for 10 h under oxygen gas flow. The NM7525 and NCM811 electrodes were fabricated by applying a slurry composed of 96 wt% NM7525 as the active material, 2 wt% conductive carbon, and 2 wt% poly(vinylidene fluoride) (PVDF)-based binder (Solef 5130) dissolved in anhydrous *N*-methyl-2-pyrrolidone (NMP) solvent (Sigma-Aldrich) on aluminum foil. Then, the electrode was pressed by a doctor blade method and dried in a vacuum at 120 °C for ≈ 1 h. All experiments were conducted with a high loading mass of 21 mg cm⁻². The graphite anodes were fabricated with the graphite and cmc (sodium salt of carboxymethyl cellulose)/sbr (styrene-butadiene rubber) binder on the Cu foil.

Electrochemical Measurements: The electrochemical performances were evaluated using the 2032 coin-types NM7525||graphite full cell. Each cell contained a polypropylene (PP) separator. They were fabricated in an Ar-filled glove box with oxygen and moisture levels below 1.0 ppm. Galvanostatic cells of charge/discharge tests for the NM7525||graphite full cell were precycled in the voltage range of 2.9–4.45 V at room temperature to activate the battery with 3 cycles. The capacity balance (N/P) of negative and positive electrodes was 1.1–1.15. Then, the rate performance was tested at various current densities (30, 60, 100, 200, 300, 500, 700, and 1000 mA g⁻¹) at 45 °C using a battery tester (WBCS3000, WonATech Co., Ltd.). Cyclic voltammetry (CV) tests were performed using an electrochemical analyzer (WBCS3000, WonATech Co., Ltd.) at different scan rates from

0.1 to 0.5 mV s⁻¹. Galvanostatic intermittent titration technique (GITT) measurement was conducted at a constant current density of 30 mA g⁻¹ with 30-min rest durations after a pre-cycle under a temperature of 45 °C.

Materials Characterization: The crystal structures of the NM7525 cathode were examined by X-ray diffraction (XRD; PANalytical) with Mo K α radiation (λ = 0.70932 Å) in the 2θ range of 4.61°–34.32° (a step size of 0.01°). Then, the angles of the obtained XRD patterns were converted through Cu K α radiation (λ = 1.54178 Å). The Rietveld refinement (Full-Prof) was conducted to analyze the XRD patterns.^[49] The morphology and microstructures of NM7525 cathode particles were observed at an accelerated voltage of 15 keV using filed-emission scanning electron microscopy (FE-SEM; ZEISS GeminiSEM 560) at the National Center for Inter-university Research Facilities (NCIRF) at Seoul National University. For the cross-sectional SEM analyses, the pristine and cycled electrode materials were prepared by polishing each particle via a cross-sectional polisher (CP; JEOL Co., Ltd.). High-resolution transmission electron microscopy (HR-TEM) combined with selected area electron diffraction (SAED) was conducted to identify the crystal phase of the NM7525 cathode particles using filed-emission transmission electron microscopy (FE-TEM; JEM-F200, JEOL Co., Ltd.) installed at the Cooperative Center for Research Facilities (CCRF) at Sungkyunkwan University. To confirm the atomic ratio of Ni and Mn elements, inductively coupled plasma atomic emission spectroscopy (ICP-AES) was performed at the National Center for Inter-university Research Facilities (NCIRF) at Seoul National University. X-ray photoelectron spectroscopy (XPS) measurements were carried out to investigate the CEI components using an AXIS Supra (Kratos Analytical) at the National Center for Inter-University Research Facilities (NCIRF) at Seoul National University. All samples were fitted using the XPSPEAK41 software and calibrated to the binding energy (284.5 eV) of C 1s peak. Attenuated total reflection-Fourier transform infrared spectroscopy (ATR-FTIR) was conducted using an IR 4700 spectrometer (JASCO) equipped with a single reflection ATR kit (ZnSe Crystal Kit) in the scanning range of 600–4000 cm⁻¹ at the Cooperative Center for Research Facilities (CCRF) at Sungkyunkwan University.

To analyze 3D volumetric data of pristine and cycled cathode electrodes, serial-section electron tomography was used using a dual-beam focused ion beam (FIB; Helios 5 HX DualBeam™). Cross-sectional SEM imaging for the whole scale of secondary cathode particles was performed to acquire 450 images during continuous FIB milling along the depth direction at a thickness interval of 15 nm. After the experimental acquisition, a commercial 3D software package (AVIZO, Thermo Fisher Scientific) was used to stack and align the sectioned images for each sample, allowing to reconstruct a 3D artificial volume. This reconstructed volume data was then processed further to perform quantitative image segmentation analysis, measuring total porosity and spatial distribution of pores and cracks in the volumes.^[24] To enhance local contrast in the fine details of cracks and pores, a contrast-limited adaptive histogram equalization (CLAHE) filter was applied to all the images.^[26] To reduce probable contamination during the serial-section imaging process, pretreatment for cleaning was conducted using Ar plasma for 20 min after loading the samples into the FIB-SEM chamber. For FIB milling, an accelerating voltage of 30 kV and a current of 2.4 nA were used. All cross-sectional images by SEM were obtained in backscatter electron (BSE) mode at a low accelerating voltage and low current of 5 kV and 0.1 nA, respectively, to reduce the effects of artificial charging. The artificial vertical pattern formed in the SEM images because of uneven vertical milling, the curtaining effect, was removed by applying fast Fourier transform (FFT) filtering to the entire SEM image stack. Due to the low yield of the BSE signal, which results in a poor signal-to-noise ratio, Gaussian filtering was applied to reduce statistical noise in each SEM BSE image. Digital segmentation, visualization, and quantification of the pore structures in the secondary cathode particles were also conducted using the commercial 3D software package (AVIZO, Thermo Fisher Scientific). An energy-dispersive X-ray spectrometer (EDX) equipped inside the FIB-SEM instrument was employed to analyze the elemental distribution of transition metals on secondary cathode particles before and after cycling.

Computational Details: The theoretical highest occupied and lowest unoccupied molecular orbital (HOMO and LUMO) energy levels of electrolyte components were identified through the density functional theory

(DFT) calculations Gaussian 16 software package.^[50] Following the calculations, spin-unrestricted density functional theory (DFT) was applied to optimize the molecular structures with no symmetry constraints using the Becke–Lee–Yang–Parr (B3LYP) hybrid exchange–correlation functional and the triple-zeta valence polarization (TZVP) basis set.^[51–56] The calculated molecular geometries of electrolyte components, such as the additives and solvents, were constructed using the Avogadro software.^[57] The dielectric constant ($\epsilon = 21.468$) was adopted, corresponding to the experimental value for the EC/EMC/DMC (2:4:4 vol%) mixture as a commonly used electrolyte solvent based on previous studies.^[58–60]

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

3D tomography, cathode, Co-free, Li-ion battery, secondary particle

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